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RiskAnalyzer (ARAPL_Configuration_V13) Data Catalog

Standalone reference to every table and field in the RiskAnalyzer database itself — the destination system that TSRe data is converted into. Same style as the TSRe catalog: organized by function, not by what any one script uses. 14 tables total, grouped into 6 functional areas.

1. Instrument Master Record — ILSDData

The largest table (110+ fields) — the master record for one ILS instrument (a CAT bond, sidecar, or reinsurance-style security). ILSInstrumentID is the primary key referenced by nearly every other table in this schema.

Identity & classification

Field	Meaning
ILSInstrumentID	Primary key (identity)
ILSLayerID, CipherID, PricingLayerID	Internal layer/cipher identifiers used to cross-reference other systems
ILSName, Issuer_Name, Class, Series	Bond identity — name, issuer, tranche class/series
COMPANY_ILS_Vehicle, SponsorID, CompanyID	Sponsor/vehicle identity
Security_or_Reinsurance_Type	Structure type (e.g., CAT Bond, sidecar, CRI, ILW) — drives downstream logic (see the earlier gap analysis for the string-matching bug tied to this field)
ParentInstrumentID, ParentILSInstrumentName	Parent/child linkage for tranches or step layers
Transaction_Description	Free-text deal description
CUSIP_ID, BB_ID, NREID, PIMCOID, IsInID	Cross-system security identifiers (CUSIP, Bloomberg, and other data-vendor IDs)

Dates & tenor

Field	Meaning
Submission_Date, Expected_Modeling_Date, CreatedDate, DateSubmissionReceived	Workflow/tracking dates
Risk_Period_Inception, Risk_Period_Expiration	Coverage period

Field	Meaning
Target_Maturity_Date, Final_Maturity_date, Early_Redemption_Date, Final_Extended_Redemption_Date	Bond maturity/redemption dates
Next_Call_Date, Next_Call_Price	Call option terms
ResetDate	Reset date for floating terms
IssuedDate	Original issuance date
Risk_Period, Tenor, RiskSeasons, RemainingRiskPeriod	Duration measures (years/seasons of risk covered, remaining term)

Currency & pricing

Field	Meaning
Currency, ExchangeRate	Denomination and FX rate
PriceInitial, PrincipalAmountInitial	Original issuance price/notional
OfferingCoupon, InitialCouponGuidance	Coupon terms at issuance
ReferenceRate, BenchmarkIndex	Floating-rate reference (e.g., SOFR)
IRR_AT_Issuance, IRR_Today	Internal rate of return at issuance vs. currently
YTM_At_Issuance, YTM_Today	Yield to maturity at issuance vs. currently
Return_On_Invested_Capital, Risk_Adjusted_Return_On_Capital	Return metrics
Reduced_Interest_Spread	Spread after any step-down/reduction
RateOnSubjectBase, AdjRate	Rate-on-line and adjusted rate figures

Modeled risk metrics (as-of-issuance / offering-circular figures)

Field	Meaning
Modeled_AAL_Base, Modeled_AAL_Sensitivity	Modeled expected loss — base case and sensitivity case
Modeled_attachProb_Base, Modeled_attachProb_Sensitivity	Modeled attachment probability — base/sensitivity
Modeled_ExhaustProb_Base, Modeled_ExhaustProb_Sensitivity	Modeled exhaustion probability — base/sensitivity

(Note: these are the static, offering-circular figures — separate from ILSResults, which holds the figures computed from the live simulation. See the earlier gap analysis on why these two are never reconciled.)

Coverage & trigger

Field	Meaning
CoveredPerils, TerritorialScope	Perils and geography covered (free text, parsed from TSRe's CAT bond attribute in the conversion script)
TriggerType, LossBasis	How the instrument is triggered (indemnity/industry-index/parametric) and loss basis
RiskModelingFirm	Modeling firm (e.g., AIR/Verisk)
PrincipalReductionType	How principal is reduced on a loss event

Field	Meaning
Exposure_Vintage	Vintage/year of the exposure data underlying the modeling
Extension_Information_I/II/III	Extension/step-down provision terms (hardcoded to 250/50/10 bps industry-wide in the current conversion script — see earlier gap analysis)

Ratings

Field	Meaning
Issuer_Credit_Rating, AMBest_Ratings, SnP_Ratings, Demotech_Ratings	Credit ratings from multiple agencies

Economics — fees, expenses, structuring costs

Field	Meaning
Initial_Fees, ManagementFee, Hurdle, Origination, UWOverride, Discount, FET, Brokerage, Setup, AcquisitionCost, FrontingFee, LeverageFeeAllocation, LeverageFee, FixedExpenses, VariableExpenses, HedgingFees, PlacementFeeNSetUpCosts	The full fee/expense stack for structuring and running the deal
CedingCommission, ProfitCommission	Ceding/profit commission terms (common in quota share / sidecar structures)
ReinstatementPercentage, ReinstatementFET	Reinstatement cost terms
Collateral, Escrow_Model, IndentureCollateral	Collateral structure

Status & workflow

Field	Meaning
Status, Data_Available, Modeling_status, Modeling_Agent	Workflow/tracking status flags
IsILSDeleted, IsBound, IsQuoted	Lifecycle flags
AuditSignOff	Sign-off tracking
COMPANY_Watchlist, Watchlist_desc	Watchlist flagging and notes
UserName	Last user to touch the record

Counterparties & sourcing

Field	Meaning
Broker, COMPANY_broker, BrokerContactEmail	Broker of record and contact
Placement_agent	Placement agent
Offering_Form	Offering document type (e.g., 144A, Reg S)
Artemis_URL	Link to a public deal-tracking source (Artemis.bm)

Misc bond mechanics

Field	Meaning
Callable_Bond, CallProvisions	Call features
Zero_Coupon_Bond	Zero-coupon flag
CouponFrequency	Payment frequency
PayoutOption	Payout mechanism
Permitted_Investment	Collateral investment restrictions
ReportingFreqForProRata	Reporting frequency for pro-rata treaties

2. Treaty/Layer Terms — ILSTerms

The RiskAnalyzer-side landing table for treaty/layer economics — the direct analog of TSRe's tReinsuranceTreaty. One row per layer/treaty, linked to ILSDData via ILSInstrumentID. This is the table arapl_populate_ilsresults reads PrincipalAmount/AggregateLimit/OccurrenceLimit/Percent from (see the gap analysis).

Field	Meaning
ILSTermsID (PK), ParentILSTermsID, ParentRelationType	Identity and parent/child layer relationship
AnalysisSID, ProgramID, ReProgramSID, ReinsuranceID	Links back to the originating TSRe analysis/program/treaty
InceptionDate, ExpirationDate, CurrencyCode	Coverage period and currency
CedantName, ReinsurerName	Counterparties
OccurrenceLimit, OccurrenceRetention, AggregateLimit, AggregateRetention, RiskLimit, RiskRetention	Attachment/limit structure — mirrors TSRe's tReinsuranceTreaty
CededAmount, PlacedAmount, CoinsuranceAmount	Cession/placement/coinsurance terms
IndustryLossMinimum/Maximum, CompanyLossMinimum/Maximum	Industry- or company-loss trigger basis
PrincipalAmount, PercentAnnualSpread	Bond notional and coupon spread
ParticipationNet, ParticipationGross	Participation share (net/gross)
Premium, PremiumMinimum, PremiumEstimated, ReinstatementPremium, LayerPremium	Premium terms
Reinstatement, ReinstProvisionalAmount, ReinstProvisionalTime, ReinstatementProvisions	Reinstatement mechanics
LossesPaid	Actual losses paid to date
LossModificationFactor, LossBasis	Loss adjustment factor and basis
TriggerTypeCode, PerilSetCode, PerilDescription	Trigger type and peril coverage (coded + description)
AppliesToAreaSID, AppliesToAreaName	Geographic scope
InuringSequenceNumber, StartingEventNumber	Layer sequencing
PreferredExposureTypeCode, ReinsTargetTypeCode, SubmitStatusCode, ReinsuranceTypeCode	Various TSRe-sourced classification codes
IsDeleted	Soft-delete flag
DeductibleType	FK to ILSDeductibleTypes
RIPBrokerage	Reinsurance-to-close/RIP brokerage cost

Fields that appear to be added specifically in RiskAnalyzer (not obviously TSRe-sourced)

Field	Meaning
100Perc_Attachment	100%-basis attachment point
Attachment_Basis	Basis the attachment is expressed on
Co-Participation	Co-participation percentage
Coverage_Type	Coverage type classification
Minimum_Net_Retention	Minimum net retention
Principal_Limit_Band	Principal limit banding
QS_Cession	Quota-share cession percentage
AggregateDeductible	Aggregate deductible amount
AdjustableRate, SwingRate	Adjustable/swing-rated premium terms
DepositPremiumHundredPercent, MinimumPremiumHundredPercent, EstSubjectBasePremium	Deposit/minimum premium terms on a 100% basis, and estimated subject-base premium

Association/junction tables

Table	Fields	Purpose
ILSTermsLossSetAssociation	ILSTermsID, LossSetID, LMF (loss modification factor), Description	Links a layer to one or more loss sets, each with its own modification factor
ILSTermsPeril	ILSTermsID, PerilID	Many-to-many: which perils apply to this layer
ILSTermsRegion	ILSTermsID, CountryID	Many-to-many: which countries/regions apply to this layer

3. Computed Risk Statistics — ILSResults

The output of `arapl_populate_ilsresults` — the actual computed risk metrics from the simulation, as opposed to the static offering-circular figures in ILSDData.

Field	Meaning
ID (PK), ILSInstrumentID, ILSTermsID ActivityID, AnalyzeReActivityID	Links back to the instrument and specific layer Links to the analysis run that produced this result
Vendor, TemplateName, Perspective	Which model vendor, template, and financial perspective this result reflects
LossType	AEP / OEP / TVAR (annual exceedance, occurrence exceedance, or tail value-at-risk)
PerilRegionID	Peril/region breakdown identifier (currently hardcoded to a single "All/All" bucket per the gap analysis)
AAL, StandardDeviation 1_111yr, 2yr, 5yr, 10yr, 25yr, 50yr, 100yr, 250yr, 500yr, 1000yr	Average annual loss and its standard deviation Loss at each return period
AttachProb, ExhaustProb	Probability of attaching / fully exhausting the layer
CipherID	Cross-reference cipher identifier

Field	Meaning
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4. Historical Event Benchmarking & Loss Adjustment

ILSHistorical

Per-event historical loss comparison for a given instrument/analysis.

Field	Meaning
ILSInstrumentID, ActivityID, HistoricalEventsID, EventID	Identity/links
Description, EventYear	Which historical event
ModeledLoss, GrossLoss	Modeled loss for this instrument from this historical event, at net and gross
ActualLoss, ActualLossOriginal	Actual reported loss (and an original/unadjusted version)
ModeledIndustryLoss	Modeled loss at the industry level for the same event
IndustryReturnPeriod, PortfolioReturnPeriod	Implied return period of the event, from an industry view and a portfolio view
Comments	Notes

ILSLossModificationFactors

Adjustment factors applied to modeled losses, by peril/region/model.

Field	Meaning
ActivityID, PerilRegionID, PerilID, CountryID, ModelID	Scope of the adjustment
ModelAdjustment	Adjustment for model uncertainty/limitations
ExposureGrowth	Adjustment for exposure growth since the model's vintage
DataQuality	Adjustment for exposure data quality
TotalAdjustment	Combined/net adjustment factor
LossAdjustmentExpense	ALAE loading
InsuranceToValue	Insurance-to-value adjustment
EventSeverityID	Severity bucket the adjustment applies to

ILSAdjustmentDetails

Free-text-heavy record of adjustment rationale, by instrument and model type.

Field	Meaning
ILSInstrumentID, ModelType	Scope
AdjType, AdjDesc	Type and description of the adjustment
EmbeddedAdjFactor, UserAdjFactor	The model's own embedded adjustment vs. a user-overridden one
Comments	Notes/rationale

5. Analysis Run Control

ILSActivityMonitor

The job-control table — one row per analysis run. ActivityID (its primary key) is the identifier threaded through every other table above.

Field	Meaning
ActivityID (PK), JobId AcctGrpId	Run identity The ILSInstrumentID this run is for (joined against ILSDData)
UserId, SubmissionDateTime, LastUpdationDateTime, StartTime, EndTime ARADbName, SourceDbName, AcctName AnalysisType, AnalysisMode, AnalysisName Vendor	Who ran it and when Source database/account context What kind of analysis this is Model vendor (AIR/RMS) — hardcoded to 'AIR' in the current conversion pipeline regardless of what's stored here
NumberOfTrials	Number of simulation trials/years — the authoritative source the conversion script <i>should</i> be reading instead of hardcoding 10,000 (see earlier gap analysis)
Perspective CurrencyCode Peril, ShapeId HurricaneTemplate, EarthquakeTemplate DemandSurge MinX, MinY, MaxX, MaxY DamageRatioFileName JobStatus IsDefaultAnalysis	Financial perspective for this run Currency Peril/shape scoping Model templates used Demand-surge flag Geographic bounding box for the run Custom damage-ratio file, if used Run status code Whether this is the default run for the instrument
PortfolioGroupId Version	Portfolio grouping Software version

ILSAnalysisOptions

Field	Meaning
ActivityID, Perils	Simple link of an analysis run to the peril(s) selected

6. Reference/Lookup Tables

ILSType

Field	Meaning
ID, ILSType AnalyzeReLayerGroup	Lookup of instrument type codes/names Cross-reference to AnalyzeRe's layer grouping convention
Description	Description

ILSDeductibleTypes

Field	Meaning
ID, DeductibleType	Lookup of deductible-type codes/names, referenced by ILSTerms.DeductibleType

ILS_Security

A **flattened, denormalized summary/export table** — one row per security with the most commonly reported fields pre-joined (Company Name, Contract Name, Treaty Name, Perils, Exp Value, Std Deviation, Attach/Exp/Exhaust Prob %, CUSIP, dates, currency, Occurrence/Aggregate Limit/Retention, Principal Amt, Spread %, Sponsor, Modeling Firm, Calculation Agent, Manager, Collateral Account, Reference Rate, Trigger Type, Cover Type, Covered Perils/Areas, both sets of Offering Risk Analysis 1/2 figures, Rating Agency, Rating). This looks like a reporting or external-distribution table rather than a live operational one — it duplicates data that otherwise lives across ILSData, ILSTerms, and ILSResults.

How the RiskAnalyzer tables relate to each other

- **ILSActivityMonitor** (one row per analysis run, keyed by ActivityID) feeds →
 - **ILSResults** (computed AAL / return-periods / AttachProb / ExhaustProb, linked via ILSInstrumentID / ILSTermsID)
- **ILSData** (instrument master record) is the parent of:
 - **ILSTerms** (layer/treaty terms), which is in turn the parent of:
 - * ILSTermsPeril
 - * ILSTermsRegion
 - * ILSTermsLossSetAssociation
 - ILSAdjustmentDetails
 - ILSHistorical (per-event historical benchmarking)
 - ILSLossModificationFactors (adjustment factors)
 - ILS_Security (flattened summary/export view, duplicating fields from the tables above)

How this compares to the TSRe catalog

The two schemas map closely, field-for-field, in the treaty/layer area — ILSTerms is essentially a flattened copy of TSRe's tReinsuranceTreaty plus a handful of RiskAnalyzer-specific additions (100Perc_Attachment, Co-Participation, QS_Cession, SwingRate, etc.). The main differences:

- **ILSData has no real TSRe equivalent** — it's a RiskAnalyzer-only table holding investor-facing/offering-circular data (ratings, fees, IRR/YTM, call provisions) that TSRe doesn't model at all. Only a slice of it (CoveredPerils, TerritorialScope, a handful of attribute-sourced fields) is actually populated from TSRe's tCatBondAttributeValue.
- **ILSResults and ILSHistorical have no TSRe source** — they're purely computed/derived in RiskAnalyzer from the simulated loss output and external historical-industry-loss data.
- **TSRe's richer structural detail doesn't fully survive the trip.** TSRe's tReinsuranceProgram (coverage-by-LOB flags, inuring structure, XML layer layout) and the 8-way participation-basis loss table have no matching level of detail anywhere in the RiskAnalyzer schema — ILSTerms only has two participation fields (ParticipationNet/Gross), and program-level LOB coverage flags don't appear at all.